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Regulating Toxic Amyloid-Beta Oligomers in Alzheimer's Disease: A Control Theory Approach

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Abstract. The accumulation of amyloid-beta (Aβ) plays a critical role in Alzheimer's disease progression. We present a mathematical framework using stochastic Smoluchowski-type models to investigate the aggregation and degradation dynamics of Aβ in the brain. The framework allows us to capture the evolution of monomers, dimers, and higher-order polymers, emphasizing their interplay under physiological and pathological conditions. To mitigate the pathological effects of Aβ aggregation, we formulate an optimal control problem incorporating control inputs designed to reduce dimer and polymer densities. The objective function minimizes polymer density over the treatment period and dimer density at the final time, reflecting immediate and long-term therapeutic goals. Numerical simulations explore the effects of key parameters on Aβ dynamics, including clearance rates, weight factors, and control input limits. Our results demonstrate the effectiveness of optimal control strategies in reducing pathological aggregates while preserving system stability.

Keywords: Alzheimer's disease, monomers, toxic oligomers, stochastic models, brain proteins, control strategies in disease treatment

1 Introduction

Alzheimer's disease (AD) is a progressive neurodegenerative disorder characterized by cognitive decline and memory loss, affecting millions of individuals worldwide [1]. It is closely associated with the accumulation of beta-amyloid (Aβ) peptides in the brain, which play a crucial role in the disease's pathology [2]. Aβ peptides are initially produced as soluble monomers by neurons but can aggregate into oligomers and fibrils, contributing to the formation of amyloid plaques, a hallmark of AD. Aβ peptides, primarily Aβ_{40} and Aβ_{42}, are known for their propensity to self-assemble into oligomers, fibrils, and insoluble plaques, contributing to neuronal toxicity [3]. Molecular dynamics simulations provide valuable insights into these peptides' conformational flexibility, binding affinities, and aggregation pathways at atomic and molecular scales.

Mathematical modelling in Alzheimer's disease provides a powerful tool to study the complex mechanisms underlying it, enabling researchers to gain insights into its progression, predict outcomes, and develop therapeutic strategies. The Smoluchowski equation is one of the fundamental mathematical models used in molecular dynamics to describe the time evolution of the probability distribution of particle positions under the influence of diffusion and external forces [4]. Originally derived to model Brownian motion, which governs the dynamics of particles in a fluid where inertial effects are negligible and stochastic thermal fluctuations dominate. In molecular dynamics, the Smoluchowski equation provides a powerful framework for studying aggregation, nucleation, and reaction kinetics, particularly in systems where interactions between particles or subpopulations lead to complex collective behaviour. This approach has been used to model the Aβ aggregation and degradation in the brain [5–7].

Aβ is a peptide derived from the amyloid precursor protein (APP) through sequential proteolytic cleavage by β-secretase and γ-secretase enzymes [8]. The molecular structure of Aβ varies depending on its aggregation state, which is critical in its pathological effects in Alzheimer's disease. Monomeric Aβ is a soluble, unstructured peptide, typically 39–43 amino acids long, with Aβ_{40} and Aβ_{42} being the most common isoforms [9]. Aβ_{42} is more hydrophobic, prone to aggregation, and strongly associated with neurotoxicity. As Aβ aggregates, it forms oligomers, protofibrils, and insoluble fibrils. These pathological protein structures, including toxic Aβ oligomers and larger immobile aggregates, play a pivotal role in disease progression by disrupting cellular functions and contributing to neuronal damage. Toxic oligomers are small, soluble protein aggregates that can diffuse through the cerebrospinal fluid (CSF) and interact with neurons, leading to synaptic dysfunction and cell death [10]. On the other hand, immobile are larger insoluble deposits that accumulate in brain tissue and serve as a reservoir for toxic species. Effective therapeutic strategies aim to reduce the burden of toxic oligomers and immobile aggregates to slow disease progressions.

Optimal control theory provides a mathematical framework for designing and analyzing treatment strategies to minimize toxic protein accumulation while balancing therapeutic efficacy and potential side effects [11]. The approach involves modelling the dynamics of protein aggregation and clearance, along with control variables such as drug dosage or immune response modulation. A typical optimal control problem in this context involves an objective function that accounts for minimizing polymer accumulation and associated costs or side effects [12]. The optimal control algorithms determine time-dependent intervention strategies that maximize therapeutic benefits. We apply this optimal control theory approach to the Smoluchowski-type models describing the polymerization and aggregation processes of Aβ. Our silico control strategy targets enhancing the clearance of toxic oligomers and immobile polymers in the brain, which can significantly advance therapeutic strategies for AD.

The rest of this paper is organized as follows. In Sect. 2, we provide a detailed description of the ordinary differential equation (ODE) models that form the basis of our analysis, outlining the aggregation and degradation dynamics of amyloid-beta molecules. This is followed by the formulation of the optimal control problem in Sect. 3, where we introduce the control variables, the objective function, and the constraints governing the optimization process. Sect. 4 delves into the parameter values used for the model, offering justifications based on biological relevance, and includes a thorough linear stability analysis of the control-free model. Numerical simulation results are presented and discussed in detail in Sect. 5, illustrating the system's behaviour under different parameter settings and control strategies. Finally, we summarize key findings in Sect. 6.

2 Mathematical Model For Dynamics of Brain Polymers in AD Progression

The progression of AD is linked to the dynamics of Aβ levels in the CSF. Amyloid proteins are initially produced as monomers by neurons and then diffuse through the CSF, with their diffusivity decreasing as their size increases. These monomers can aggregate and polymerize, forming longer polymer chains, and the Smoluchowski equation can be applied in this case [6]. For $n \in \mathbb{N}$, let P_n represent a polymer of length n, composed of n identical monomer units. When polymers are sufficiently close, they may merge to form a single polymer whose length equals the sum of the lengths of the two merging polymers (only binary reactions are considered). The merge of a polymer of size i with a polymer of size j can be expressed as:

$$P_i + P_j \rightarrow P_{i+j}.$$

Now, we assume that a number N exists such that polymers of length less than N are soluble and greater than or equal to N are immobile. We denote $w_n(t)$ as the number of soluble amyloid polymers of length n ($n < N$) contained in a given representative elementary volumes (REV) at time t, while $W_N(t)$ denotes the total number of immobile particles in REV. In this configuration, the aggregation rate $A_{i,j}$ of two soluble polymers of length i and j can be expressed as [6]:

$$A_{i,j} = \begin{cases} rw_iw_j & \text{if } i,j < N, i \neq j, \\ \frac{1}{2}rw_i^2 & \text{if } i,j < N, i = j. \end{cases}$$

Additionally, we assume that two immobile polymers do not merge, and the rate at which a soluble and an immobile polymer aggregate is given by $A_{i,N} = A_{N,i} = r_s w_i w_N$ if $i < N$. Furthermore, we assume $r_s < r$ since two soluble polymers are more likely to merge than if one is immobile. Considering these factors, the following differential equation describes the evolution of the number of monomers in Ω:

$$\frac{dw_1}{dt} = -rw_1 \sum_{j=1}^{N-1} w_j - r_s w_N - m_1 w_1 + p_1, \tag{1}$$

where m_1 is the mortality rate due to the clearance activity by microglia and other blood vessels [13, 14], and p_1 is the production rate of monomers. Similarly, the evolution of the number of polymers of length n $(n < N)$ can be described by

$$\frac{dw_n}{dt} = \frac{r}{2} \sum_{i+j=n} w_i w_j - r w_n \sum_{j=1}^{N-1} w_j - r_s w_n w_N - m_n w_n, \tag{2}$$

where m_n is the mortality. Finally, the evolution of the number of immobile polymers can be described by

$$\frac{dw_N}{dt} = \frac{r}{2} \sum_{\substack{i,j<N \\ i+j\geq N}} w_i w_j - m_N w_N, \tag{3}$$

where m_N is the clearance rate. All parameters in the system of differential equations (1)-(3) are assumed to be positive, with non-negative initial conditions.

A large number of subpopulations can be included in the analysis, but this involves more complex notation and a slight increase in computational time. However, this study focuses on a simplified version of the model by considering three subpopulations: monomers, toxic oligomers (dimers), and immobile aggregates (polymers). Here, we rescale w_1, w_2, and w_3 by

$$x(t) = \frac{w_1(t)}{K}, \quad y(t) = \frac{w_2(t)}{K}, \quad z(t) = \frac{w_3(t)}{K},$$

where K is the number of monomers in a nanogram of amyloid-beta. In this setting, the system of equations (1)-(3) can be transformed into

$$\begin{aligned} \frac{dx}{dt} &= -kx^2 - kxy - k_s xz - m_1 x + p, \\ \frac{dy}{dt} &= \frac{1}{2}kx^2 - kxy - ky^2 - k_s yz - m_2 y, \\ \frac{dz}{dt} &= \frac{1}{2}ky^2 + kxy - m_3 z, \end{aligned} \tag{4}$$

where $k = rK$, $k_s = r_s K$, and $p = p_1/K$. This system is considered under non-negative initial conditions.

3 Optimal Control for Mitigating Pathological Effects of Aβ Aggregation

The optimal control problem is a function that minimizes the dimer and oligomer numbers in the brain. Besides, the costs associated with implementing an optimal control strategy have also been added. The function is defined over a time period of $[0, T]$. The first controller applied to the model is control input u_1, which indicates the strength of the clearance of dimer amyloid-beta in the brain. Moreover, the second controller applied to the model is control input u_2 to clear the oligomer in the brain. Both of these control functions optimize the optimal strategy in drug development. Including these controllers in the dynamics of the model (4), it becomes

$$\begin{aligned}
\frac{dx}{dt} &= -kx^2 - kxy - k_s xz - m_1 x + p,\\
\frac{dy}{dt} &= \frac{1}{2}kx^2 - kxy - ky^2 - k_s yz - m_2 y - u_1 y,\\
\frac{dz}{dt} &= \frac{1}{2}ky^2 + kxy - m_3 z - u_2 z,
\end{aligned}$$

with the optimal control problem of the form

$$\min_{(u_1,u_2)\in\mathcal{U}} J(u_1(t),u_2(t)) := \frac{1}{2}\int_0^T (b_1u_1^2 + b_2u_2^2 + c_1z^2)dt + c_2y(T), \qquad (5)$$

where $\mathcal{U} = \{(u_1,u_2) \in (L^\infty(0,T))^2 : 0 \le u_1(t) \le u_1^{\max}, 0 \le u_2(t) \le u_2^{\max}\}$. The parameter c_1 is a positive weight factor that helps to achieve balance in the optimization function. It describes how much factor is given to the portion of the polymers that causes an imbalance of the amyloid-beta in the brain. The last term in the optimization function ensures the total portion of dimers at the final time T with the weight factor c_2. The factors b_1 and b_2 measure the relative cost of optimal control input and can be adjusted as desired.

Now, we derive the first-order necessary conditions for optimal control by constructing the Hamiltonian (H) and applying Pontryagin's maximum principle [15, 16]. We denote $\mathbf{x}(t) = (x(t),y(t),z(t))^T$, $\mathbf{u}(t) = (u_1(t),u_2(t))^T$, and $\Lambda(t) = (\lambda_1(t),\lambda_2(t),\lambda_3(t))^T$. With these notations and the controller (5), we define the Hamiltonian as

$$\begin{aligned}
H = \frac{1}{2}(b_1u_1^2 + b_2u_2^2 + c_1z^2) &+ \lambda_1(-kx^2 - kxy - k_sxz - m_1x + p)\\
&+ \lambda_2((k/2)x^2 - kxy - ky^2 - k_syz - m_2y - u_1y)\\
&+ \lambda_3((k/2)y^2 + kxy - m_3z - u_2z).
\end{aligned}$$

Suppose $\mathbf{u}^* = (u_1^*, u_2^*)^T$ is the optimal control and $\mathbf{x}^*(t) = (x^*(t), y^*(t), z^*(t))^T$ is the corresponding optimal trajectory. Then, the optimality conditions are given by

$$\left.\frac{\partial H}{\partial u_1}\right|_{\mathbf{u}=\mathbf{u}^*} = 0 \quad \text{and} \quad \left.\frac{\partial H}{\partial u_2}\right|_{\mathbf{u}=\mathbf{u}^*} = 0.$$

After simplification of these equations, we obtain

$$\begin{aligned}
u_1^*(t) &= \min\left\{u_1^{\max}, \max\left\{0, \frac{\lambda_2(t)y(t)}{b_1}\right\}\right\},\\
\text{and}\quad u_2^*(t) &= \min\left\{u_2^{\max}, \max\left\{0, \frac{\lambda_3(t)z(t)}{b_2}\right\}\right\}.
\end{aligned}$$

In addition, the adjoint system leads to

$$\begin{aligned}
\frac{d\lambda_1}{dt} &= (2\lambda_1 - \lambda_2)kx + (\lambda_1 + \lambda_2 - \lambda_3)ky + \lambda_1(k_s z + m_1), \\
\frac{d\lambda_2}{dt} &= (\lambda_1 + \lambda_2 - \lambda_3)kx + (2\lambda_2 - \lambda_3)ky + \lambda_2(k_s z + m_2 + u_1), \\
\frac{d\lambda_3}{dt} &= k_s(\lambda_1 x + \lambda_2 y) + \lambda_3(m_3 + u_2) - c_1 z,
\end{aligned}$$

and the transversality conditions

$$\lambda_1(T) = 0, \quad \lambda_2(T) = c_2, \quad \text{and} \quad \lambda_3(T) = 0.$$

Since the state equations have initial conditions coupled with the adjoint equations, which in turn have final time conditions, we use an iterative method known as the forward-backwards sweep algorithm to solve the system [17].

4 Parameter Setup and Linear Stability Analysis

This section outlines the assumptions made for the various parameters. We set $K = 10^{11}$ based on the fact that the mass of an amyloid-beta monomer is approximately 8×10^{-11} nanograms [6, 18]. The monomer production rate depends on the total number of REVs used to represent the brain, which we assume to be around 500, all of equal size. Setting $p = 2$ corresponds to a daily monomer production of 2 nanograms per REV, totalling 1000 nanograms for a healthy brain [6]. While explicit formulas for k and k_s are controversial [19, 20], however, we adopt the values $k = 10^{-4}$ and $k_s = 5 \times 10^{-6}$ from [6]. The parameters involved in the control problem are chosen as bifurcation parameters, as our goal is to demonstrate the possible outcomes of the model in the presence and absence of optimal control.

Stability theory is crucial in nonlinear dynamics related to disease progression and treatment. It forms the basis for predicting outcomes, planning effective treatments, and optimizing therapeutic strategies, enhancing results for individuals with neurological disorders and other dynamic health challenges. Linear stability analysis is a mathematical method used to assess the stability of equilibrium points in ODE models. This approach involves linearizing the nonlinear ODE system around an equilibrium point (x_*, y_*, z_*) and analyzing the eigenvalues of the resulting linear system. These eigenvalues determine whether deviations from the equilibrium will amplify or diminish. The equilibrium point(s) of the control-free model (4) represent its steady states and are determined as the solutions of the corresponding system of algebraic equations:

$$\begin{aligned}
-kx^2 - kxy - k_s xz - m_1 x + p &= 0, \\
\frac{1}{2}kx^2 - kxy - ky^2 - k_s yz - m_2 y &= 0, \\
\frac{1}{2}ky^2 + kxy - m_3 z &= 0.
\end{aligned}$$

The nonlinearity of this system makes finding exact solutions challenging. Therefore, we determine the solutions numerically by using the parameter values specified for the model. Numerical methods provide a robust approach to understanding the system's behaviour when analytical solutions are infeasible due to complexity. The stability behaviours of an equilibrium point can be analyzed by examining the eigenvalues of the Jacobian matrix evaluated at that point. If the real parts of all the eigenvalues are negative, the equilibrium point is stable, indicating that small perturbations around the point will decay over time, and the system will return to equilibrium. Conversely, if any eigenvalue has a positive real part, the equilibrium is unstable, and small perturbations will grow, driving the system away from the equilibrium.

5 Numerical Simulations and Discussions

This section illustrates the silico trials for optimizing control inputs u_1 and u_2. In these trials, control inputs are switched off and on to analyze their influence on AD progression. Before exploring the optimal control settings, we examine the long-term dynamics of monomers, dimers, and polymers in the brain. Figure 1 illustrates these dynamics for varying polymer clearance rates. In the first scenario, all molecular species share the same clearance rate $m_1 = m_2 = m_3 = 0.01$, with the results depicted by the solid curves in the upper panel. The solution of the ODE model converges to its equilibrium point $(x_*, y_*, z_*) = (90.23, 20.26, 20.94)$. Monomers, being smaller and simpler, are more readily cleared by cellular mechanisms, whereas dimers and polymers tend to resist degradation and clearance due to their larger size and structural complexity. We consider different clearance rates for dimers and polymers to investigate these effects. Two cases are simulated: one where dimers and polymers share the same clearance rate and another where the polymer clearance rate is reduced to half that of the dimers. In Fig. 1, the middle panel is for $m_1 = 0.01, m_2 = m_3 = 0.005$, and the lower panel is for $m_1 = 0.01, m_2 = 0.005, m_3 = 0.0025$. A substantial increase in polymer accumulation is observed when clearance rates are low, highlighting a potential risk of amyloid-beta plaque formation in the brain. In all scenarios, the initial conditions ($x_0 = 40$, $y_0 = 15$, and $z_0 = 10$) remain the same, but the curves indicate that the ODE system attains its equilibrium values within a few years, even when the initial conditions are far from equilibrium. However, in the last situation, where each substance has a different clearance rate, the results align with the AD condition because the plaque formation is more pronounced here. We consider this parameter set ($m_1 = 0.01, m_2 = 0.005$, and $m_3 = 0.0025$) for the clearance rate for the rest of the paper.

Next, we focus on the optimal control problem, where we control the dimers and polymer densities through the control inputs u_1 and u_2, respectively. The purpose is to optimize the control inputs both independently and simultaneously. According to the optimal configurations, the control input u_2 affects the polymer density, and it can take the maximum value as $u_2^{\max}$. In this setup, multiple scenarios are created in which weight factor c_1 changes continuously so that

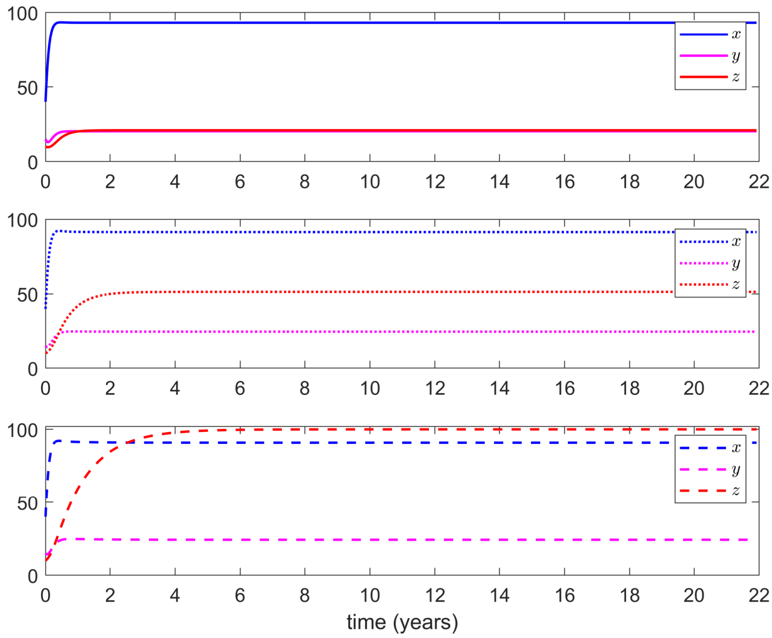

Fig. 1: (Color online) Solutions for the ODE model for different clearance rates of the monomers (blue), dimers (magenta), and polymers (red). The upper panel is for $m_1 = m_2 = m_3 = 0.01$, middle panel is for $m_1 = 0.01, m_2 = m_3 = 0.005$, and lower panel is for $m_1 = 0.01, m_2 = 0.005, m_3 = 0.0025$.

different effects on the optimal control can be analyzed. As we are interested in the control input u_2, we fix the upper limit for the control input u_1 as a positive negligible quantity such that there will be no contribution from it, and the associated cost is considered as zero. Here, we choose $u_1^{\max} = 10^{-15}$. In addition, we have fixed the parameter values $b_1 = b_2 = 1$, and $u_2^{\max} = 0.1$. Furthermore, we keep these fixed parameter values of b_1 and b_2 throughout the manuscript. Figure 2 depicts the densities of the considered substances for different values of the weight factor c_1 with no inputs from the weight associated with the dimers, i.e., $c_2 = 0$. Additionally, the initial conditions remain the same for all the simulations. It shows that the density of polymers decreases with increasing the weight factor c_1, and the opposite scenario happens in the case of monomers and dimers. We have presented the corresponding control input effort while varying c_1 in Fig. 3. This control input plot shows an increase in effort after applying the control when the weight factor c_1 is less. However, almost steady control requires a higher weight factor. Figure 2 also shows that the weight factor $c_1 = 10^{-5}$ is favourable in controlling the polymer density in the brain. In this simulation, we have fixed the upper limit of the control input as $u_2^{\max} = 0.1$, but comparing

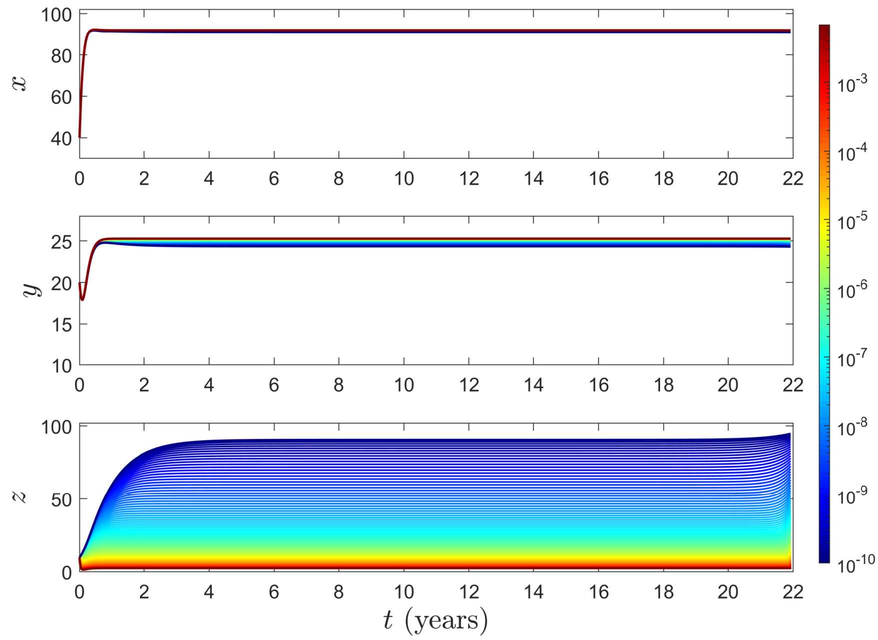

Fig. 2: (Color online) Solutions of the monomers, dimers, and polymers for the control problem for different weight parameter values of c_1 while $c_2 = 0$. The colorbar represents the values of c_1 with the other fixed values $m_1 = 0.01, m_2 = 0.005, m_3 = 0.0025$.

Figs. 2 and 3, we see that such a high upper limit does not require the control of the polymers in the brain.

We have studied the effects of the upper limit of the control input $u_2^{\max}$ in controlling the polymer density by setting the weight factor $c_1 = 10^{-5}$. The colorbar in Fig. 4 represents the range of the parameter values of $u_2^{\max}$ and we fix the upper limit for u_1 as $u_1^{\max} = 10^{-15}$. The plots in Fig. 4 show that the parameter $u_2^{\max}$ plays a crucial role in controlling the polymer density. In contrast, the densities of the monomers and dimers increase with increasing $u_2^{\max}$. Notably, the influence of $u_2^{\max}$ on the monomer density is minimal, suggesting that the monomer population remains relatively stable under variations. Moreover, it does not influence the monomer density much. Furthermore, the maximum value for $u_2^{\max}$ is considered as 14.70, but the control input u_2 does not exceed 0.03 [see the lower panel in Fig. 4]. This discrepancy underscores the efficiency of the control strategy, as even a low-intensity input $u_2^{\max}$ can effectively regulate the system. These findings highlight the nuanced role of $u_2^{\max}$ in modulating polymer dynamics while maintaining system stability.

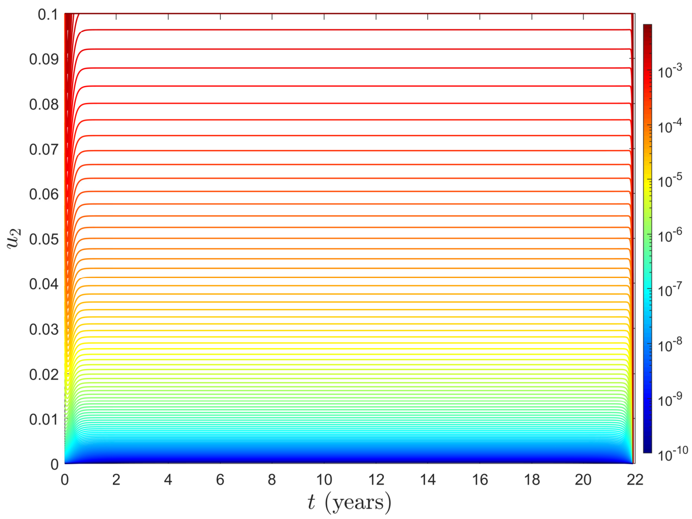

Fig. 3: (Color online) Plot of the control input u_2 while changing c_1 with $c_2 = 0$. The colorbar represents the values of c_1. The other parameters are the same as of Fig. 2.

So far, we have examined the scenario where only the input u_2 is controlled. Figure 5 depicts the solution of the control problem when the upper limit of the control input $u_1^{\max}$ varies. In this setup, we fix $u_2^{\max} = 10^{-15}$ and $c_2 = 0$ to isolate and examine the impacts of the control input $u_1^{\max}$. The colorbar in Fig. 5 represents the range of $u_1^{\max}$, and the results indicate that $u_1^{\max}$ effectively reduces significant amounts of the dimer and polymer densities. This demonstrates the critical role of $u_1^{\max}$ in mitigating the buildup of higher-order complexes within the system. On the other hand, the density of the monomers increases with $u_1^{\max}$, suggesting a redistribution of material from higher-order complexes to simpler monomers. While the increase in monomer density is not explicitly shown in the diagram, it highlights a trade-off between reducing complex densities and preserving the monomer density. These findings emphasize the utility of $u_1^{\max}$ in achieving targeted control of system dynamics, particularly in applications requiring precise regulation of complex densities.

Finally, we have presented the impact of the weight factor c_2 on the control problem in Fig. 6. In the plot, the colorbar shows the parametric range of c_2 with other fixed parameter values $c_1 = 10^{-5}$ and $u_1^{\max} = u_2^{\max} = 10^{-3}$. The results demonstrate that the densities of dimers and polymers decrease progressively as c_2 decreases, highlighting the significant influence of this weight factor on the system's dynamics. Additionally, the results suggest a potential threshold effect, where further reductions in c_2 may not yield substantial changes in the densities. As c_2 becomes smaller, the control inputs prioritize the reduction

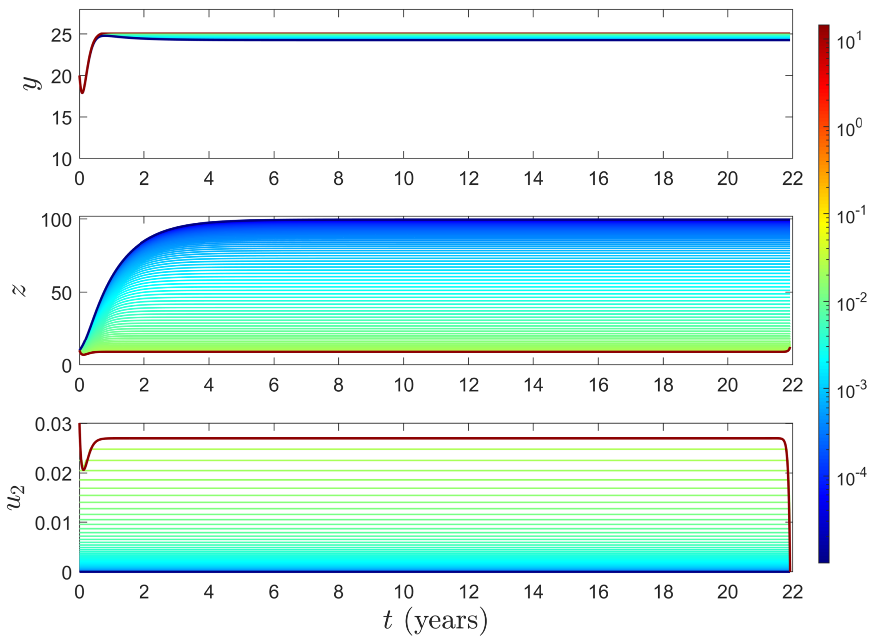

Fig. 4: (Color online) Solutions of the dimers, polymers, and the control input u_2 for the control problem for different $u_2^{\max}$ with $c_1 = 10^{-5}$ and $c_2 = 0$. The colorbar represents the parameter values of $u_2^{\max}$.

of dimer and polymer densities, potentially at the expense of less emphasis on other objectives. This highlights the need for careful tuning of c_2 to optimize system performance while avoiding diminishing returns in control efficacy. Nevertheless, these parameters can be aligned with experimental results to predict the long-term outcomes of the model. A worthwhile extension of this work in the future would be to estimate the parameters using ADNI data and clinically validated biomarkers [21]. Furthermore, employing machine learning techniques to optimize parameter estimation could enhance predictive accuracy and reveal hidden patterns within the data [22].

6 Conclusions

This work considers the molecular structure of Aβ development in the brain, which is a critical factor in the progression of Alzheimer's disease (AD). A well-known Smoluchowski equation is applied to study the aggregation and degradation dynamics of Aβ molecules, offering a mathematical framework to capture the complex interactions between these processes. The evolution of monomers,

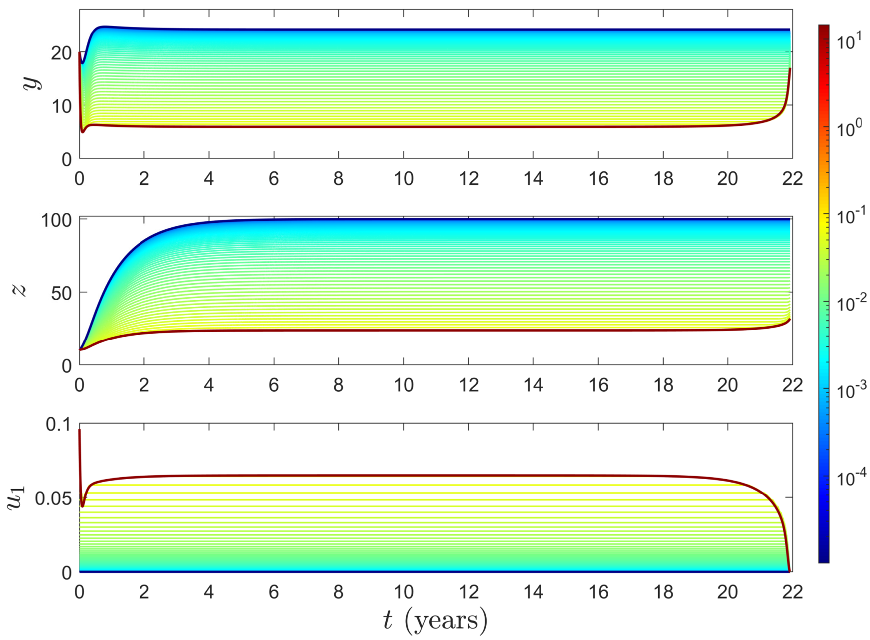

Fig. 5: (Color online) Solutions of the dimers, polymers, and the control input u_1 for the control problem for different $u_1^{\max}$ with $c_1 = 10^{-5}$ and $c_2 = 0$. The colorbar represents the parameter values of $u_1^{\max}$.

dimers, and polymers is analyzed under various circumstances, such as different clearance rates, reflecting the brain's capability to manage Aβ accumulation under normal and pathological conditions. Additionally, an optimal control problem is formulated to explore therapeutic interventions to reduce the dimer and polymer densities in the brain. The control inputs represent potential mechanisms, such as pharmacological treatments, that facilitate the clearance of these aggregates. The objective function is designed to minimize the polymer density throughout the treatment period, reflecting a sustained effort to prevent plaque formation while ensuring the reduction of dimer density at the final time, emphasizing long-term effectiveness.

We have investigated the impact of different parameters on regulating polymer density in a system driven by amyloid-beta aggregation dynamics. One key aspect is the role of the weight factor linked to the control input responsible for clearing polymers from the brain. While a higher weight effectively reduces polymer density, it minimally influences the densities of monomers and dimers. Next, the impact of the upper limit of the control input $u_2^{\max}$ is analyzed, revealing its crucial role in reducing polymer density, although monomer and dimer densities increase with higher $u_2^{\max}$. Despite higher $u_2^{\max}$ values, the actual input $u_2^{\max}$

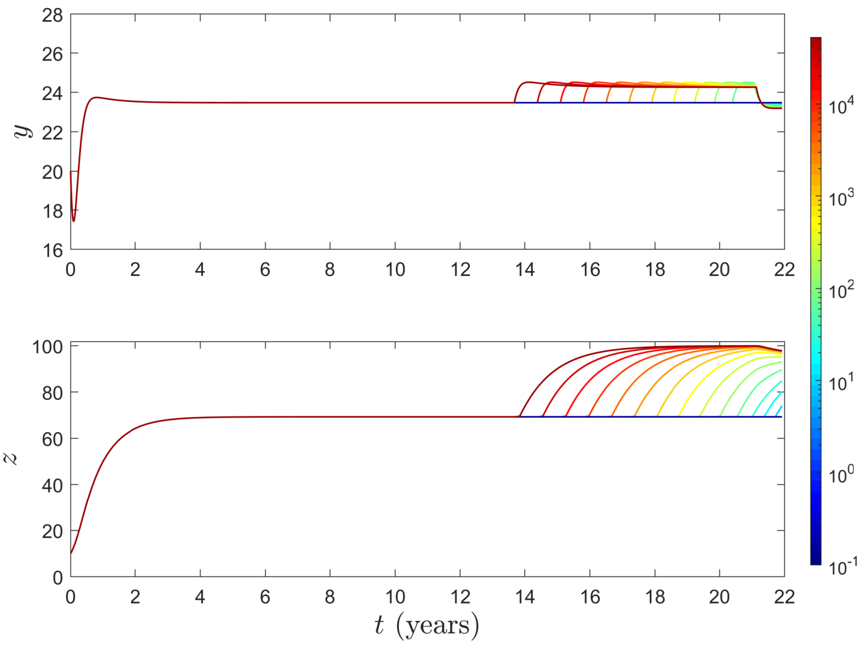

Fig. 6: (Color online) Solutions of the dimers and polymers for the control problem for different values of c_2 with $c_1 = 10^{-5}$. The colorbar represents the parameter values of c_2.

remains low for the optimal problem. Next, the focus shifts to scenarios where only u_1 is controlled. The results demonstrate that increasing $u_1^{\max}$ significantly reduces dimer and polymer densities while monomer density rises. Finally, the effect of the weight factor c_2 is explored, showing that decreasing c_2 leads to reduced dimer and polymer densities, underscoring its importance in optimizing the control strategy. These silico-trial findings highlight the interplay of control parameters in managing amyloid-beta aggregation and suggest potential avenues for therapeutic intervention.

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References

1. Monteiro, A.R., et al.: Alzheimer's disease: Insights and new prospects in disease pathophysiology, biomarkers and disease-modifying drugs. Biochemical Pharmacology 211, 115522 (2023).
2. Hampel, H., et al.: The amyloid-β pathway in Alzheimer's disease. Molecular psychiatry 26, 5481-5503 (2021).
3. Festa, G., et al.: Aggregation States of Aβ_{1-40}, Aβ_{1-42} and Aβp_{3-42} Amyloid Beta Peptides: A SANS Study. International journal of molecular sciences 20, 4126 (2019)
4. Smoluchowski, M.V.: Drei vortrage uber diffusion, brownsche bewegung und koagulation von kolloidteilchen. Zeitschrift fur Physik 17, 557-585 (1916).
5. Shaheen, H., et al.: Astrocytic clearance and fragmentation of toxic proteins in Alzheimer's disease on large-scale brain networks. Physica D: Nonlinear Phenomena 454, 133839 (2023).
6. Bertsch, M., et al.: The amyloid cascade hypothesis and Alzheimer's disease: a mathematical model. European Journal of Applied Mathematics 32(5), 749-768 (2021).
7. Ionel, C., et al.: Aβ-protein polymerization in Alzheimer disease: Optimal control for nucleation parameter estimation. Journal of Theoretical Biology 589, 111814 (2024).
8. Xu, X.: γ-Secretase catalyzes sequential cleavages of the AβPP transmembrane domain. Journal of Alzheimer's Disease 16(2), 211-224 (2009).
9. Munishkina, L.A., Fink, A.L.: Fluorescence as a method to reveal structures and membrane-interactions of amyloidogenic proteins. Biochimica et Biophysica Acta (BBA)-Biomembranes 1768(8), 1862-1885 (2007).
10. Murphy, M.P., LeVine, H.: Alzheimer's disease and the amyloid-beta peptide. Journal of Alzheimer's disease: JAD 19(1), 311–323 (2010).
11. Moore, H.: How to mathematically optimize drug regimens using optimal control. Journal of pharmacokinetics and pharmacodynamics 45(1), 127-137 (2018).
12. Joshi, H.R., et al.: Optimal control methods applied to disease models. Contemporary Mathematics 410, 187-208 (2006).
13. Colonna, M., Butovsky, O.: Microglia Function in the Central Nervous System During Health and Neurodegeneration. Annu Rev Immunol. 35, 441-468 (2017).
14. Xu, Y., et al.: Microglia in neurodegenerative diseases. Neural Regen Res. 16(2), 270–280 (2021).
15. Pontryagin, L.S., et al.: The Mathematical Theory of Optimal Processes. Wiley (1962).
16. Lenhart, S., Workman, J.: Optimal control applied to biological models. Chapman and Hall/CRC (2007).
17. McAsey, M., et al.: Convergence of the forward-backward sweep method in optimal control. Computational Optimization and Applications 53, 207-226 (2012).
18. Nag, S., et al.: Nature of the amyloid-β monomer and the monomer oligomer equilibrium. J Biol Chem. 286(16), 13827–13833 (2011).
19. Murphy, R.M., Pallitto, M.M.: Probing the kinetics of β-amyloid self-association. J. Struct. Biol. 130(2–3), 109–122 (2000).
20. Gillam, J., MacPhee, C.: Modelling amyloid fibril formation kinetics: mechanisms of nucleation and growth. J. Phys. Condens. Matter 25(37), 373101 (2013).
21. Hao, W., et al.: Optimal anti-amyloid-beta therapy for Alzheimer's disease via a personalized mathematical model. PLoS Computational Biology 18(9), e1010481 (2022).
22. Chang, C.H., et al.: Machine learning and novel biomarkers for the diagnosis of Alzheimer's disease. International Journal of Molecular Sciences 22(5), 2761 (2021).